Correlation Neighborhoods & Pairs Trading in CryptAlyst

Inverted-pair sensitivity, correction trades, and where the structure lives in the lake / refdata architecture

Design memo — for thinking, review, and staging the build

KIMI · build lane · 21 October 2026 · v1.0

Companion documents: docs/solvers.md (Stage-1 methodology), docs/postmortems/ (Stage-0 autopsy), docs/refdata.md (registry), DATA_STORAGE.md (stores census). Rules: AGENTS.md A1–A6; COMMS_PROTOCOL Rules 1 and 5.

1 Motivation: the equity analogy, in crypto

In equities, securities respond to shared events through their sector and factor structure: when a chip maker releases a new product its own price rises while close competitors sag — an **inverted pair** driven by one event. The co-movement is not anecdotal; it is the residual output of a common-factor model, and it is tradeable as relative value.

Crypto exhibits the same structure, more strongly and more measurably. Coins share common features in a disparate way: every alt perpetual loads on BTC; L1s load on each other; DeFi tokens share protocol-risk exposure; AI-themed coins share a narrative factor; memes share liquidity regime. An event on one symbol — a listing, an exploit, a funding dislocation, a regulatory headline on a category — propagates to its neighbours with measurable sign, size, and delay. That propagation graph **is** the correlation neighbourhood the registry viewer already sketches: today it shows static links (identifiers, venues, documents); this memo designs its tradeable layer.

Why now. Two system facts make this the right next move. First, the registry just became rich enough to support it: 1,396 Binance instruments (567 USDT-M perps, 828 fiat/stablecoin-quoted spot pairs) with display names, categories, supply, funding, and long/short sentiment, all keyed by canonical id. Second, the Stage-0 autopsy and Stage-1 counterfactuals show the selection-level landscape is flat — trip selection cannot pay the 20 bps round trip against a near-zero gross expectancy. The repair program therefore has to find **gross edge**, and relative-value structure is the most promising untested source: it is where systematic, fee-aware, hedged trades live.

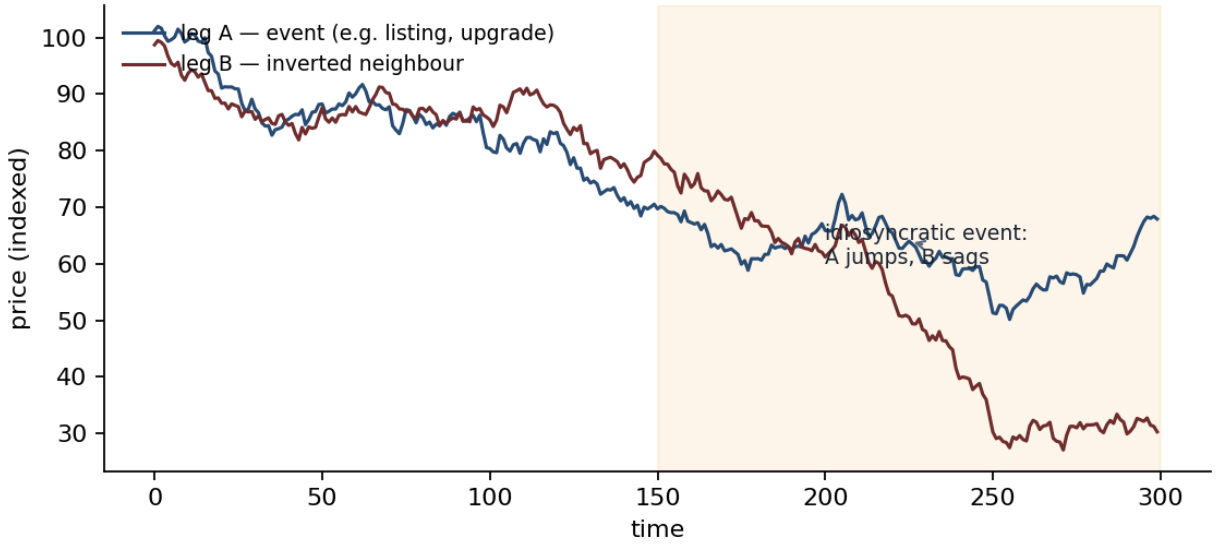


Figure 1 — Inverted-pair behaviour around an idiosyncratic event: leg A jumps on its own news, the inverted neighbour bleeds off the shared factor.

2 Concepts and formalism

2.1 The spread and its half-life

A **pair** is two instruments whose log prices share a stochastic trend. The cointegrating regression fixes the hedge ratio β, and the residual spread is the tradeable object [1][2]:

$$s_t = \ln P_t^A - \beta \ln P_t^B - \alpha, \qquad \beta \text{ from the cointegrating regression}$$

Mean reversion speed is what makes a pair worth trading. Modelling the spread as an AR(1) process gives a closed-form half-life — the expected time for half the dislocation to close — which disciplines both sizing and holding period:

$$\Delta s_t = \lambda s_{t-1} + \varepsilon_t, \qquad \tau_{1/2} = \frac{-\ln 2}{\ln(1+\lambda)}$$

Screening rule (proposed): admit a pair only when |ρ| of 5-minute returns over the estimation window is above 0.6 *and* the half-life lies between 30 minutes and 5 days. Fast half-lives are fee fodder at 20 bps round trip; slow ones tie capital and drift out of cointegration.

2.2 Factor view: where the correction trade comes from

Pairs are a two-name special case of a factor model. Every instrument's return decomposes into a market leg, a sector leg, and an idiosyncratic residual:

$$r_i(t) = \alpha_i + \beta_i^{BTC} r_{BTC}(t) + \beta_i^{sec} r_{sec}(t) + \varepsilon_i(t)$$

The **correction trade** is the trade of the residual. Concretely: an event moves leg A beyond what its factor loadings explain; the neighbourhood graph says which neighbours should absorb the spillover and with what sign; the position is long the under-reacted neighbour (or short the over-reacted one) with the factor legs hedged, and it is exited as the residual mean-reverts. This is pairs trading in the Gatev–Rouwenhorst sense [3], generalised from two names to a neighbourhood, and it is the equities literature's statistical-arbitrage practice transferred to a market where the factor structure is younger, faster, and far better instrumented [4][5].

2.3 The neighbourhood graph

Generalising from pairs to a graph: nodes are canonical instruments; edges carry typed sensitivity. Six predicates exist today (same_as, listed_on, has_document, issued_by, tracks, derived_from). The design adds four minted predicates for tradeable structure, with provenance in the edge Meta map:

Table 1 — Proposed minted predicates for tradeable structure.

predicate	meaning	edge Meta (provenance)
trades_with	co-moving legs; residual mean-reverts	window, horizon, ρ at mint, half-life
inverts_with	negative-sensitivity neighbour (equity-style inverted pair)	window, horizon, ρ at mint, event tags

same_sector	shared factor membership (L1, DeFi, AI, meme, …)	factor name, loadings, regime tag
leads_lag	one leg systematically moves first by k bars	lag k, lead-lag correlation, window

Edges are **stable structure**: minted by the estimation job, rewritten only when the estimate shifts materially (regime change), and versioned in the journal like everything else. The coefficient *series* — the thing that changes every bar — is never written to the registry; it lives in the analytics plane (§3).

3 Where the structure lives: identity vs analytics plane

The governing rule, already circulated to the build lane: **if it changes every bar it is analytics; if it changes when the exchange reconfigures it is reference.** The registry (identity plane) answers what an instrument is; the lake stores (analytics plane) hold what it did. Conflating the two would bloat the journal with per-bar noise and starve the models of history.

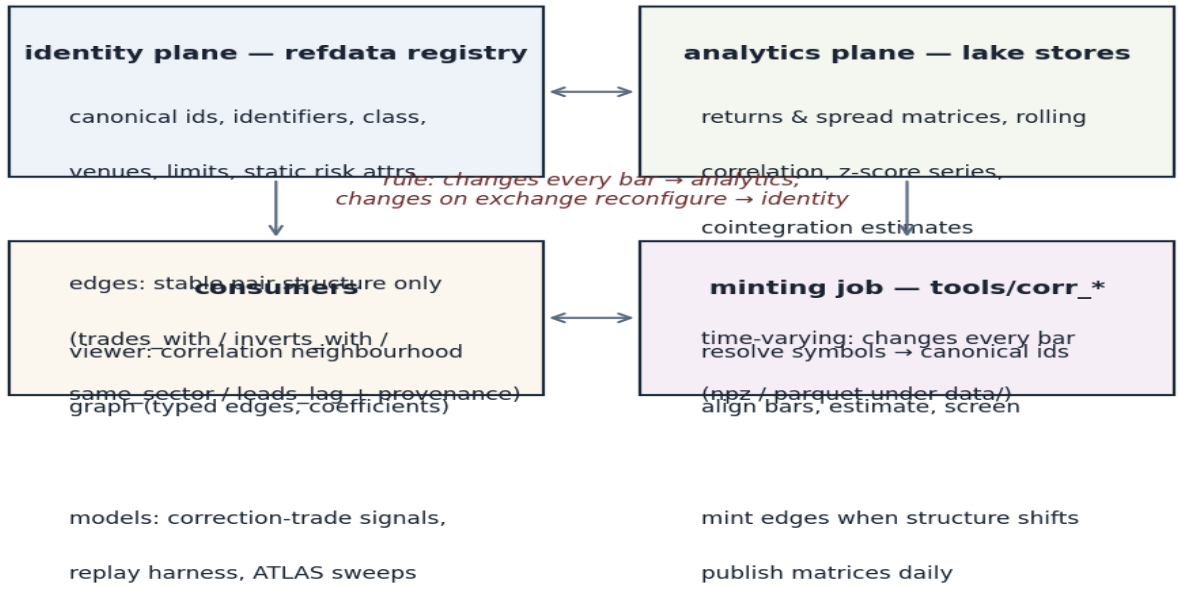


Figure 2 — Plane separation and flow. The minting job resolves symbols to canonical ids, estimates structure, mints stable edges into refdata, and publishes time-varying matrices into the lake.

Table 2 — What lives where, under the plane rule.

data	plane	store	owner
canonical id, identifiers, venues, limits	identity	refdata journal	refdata service
static risk attrs (maint margin, fee, leverage cap)	identity	refdata meta/limits	refseed generator
pair edges + provenance	identity	refdata edges	corr minting job
return matrices, ρ(t), z-scores, spreads	analytics	npz/parquet under data/	lake tools
replay counterfactuals, sweep results	analytics	docs/postmortems/	Stage-1/2 harness

4 Data, estimation, and compute plan

4.1 Inputs and alignment

Market data comes from the recorder stores; instrument identity comes from the enriched registry. The first pipeline step is **resolution**: every raw symbol in a market-data frame is resolved through the refdata client to its canonical id before alignment. This is not bureaucracy — the registry holds BTCUSDT spot and BTCUSDT perp as distinct records, and a spread computed across that seam is garbage. Legs are aligned on bar timestamps per venue session, with the exchange-native tick/lot sizes (now in every record's limits) applied before any spread is tradable-sized.

4.2 Estimation cadence

Hourly returns matrix refresh; daily full re-estimate (ρ at 5m/1h/1d horizons, cointegration screen over the perp universe, half-life filter, event-tagged inversions from funding and sentiment extremes already seeded in meta). Edge minting is *diff-driven*: only material structure changes append to the journal, so the registry stays lean while the analytics plane accumulates full history.

4.3 Compute budget and the GPU gate

A full 567×567 correlation matrix on 5-minute returns is seconds on CPU — the CPU timing baselines from Stage 1 (315 full-table passes per second over 809k rows) leave no doubt. The GPU lane (ATLAS) stays gated behind the path-level replay timing experiment per Rule 5: vectorise in numpy first, time one window, and move the hot loop to JAX/CuPy only if the measured sweep count demands it. Seeding and versioning are mandatory — a backtest that cannot be re-run identically is a liability.

5 Trading integration: the correction trade

Signal chain: (1) an event or dislocation is detected on leg A — idiosyncratic move beyond its factor expectation, funding spike, sentiment extreme; (2) the neighbourhood graph returns A's typed neighbours with sign and provenance; (3) for each admissible pair the spread z-score is computed from the analytics plane; (4) entry when |z| crosses the band, exit at the inner band, stop at k·σ beyond entry; (5) factor legs hedged so the position carries residual risk, not market risk.

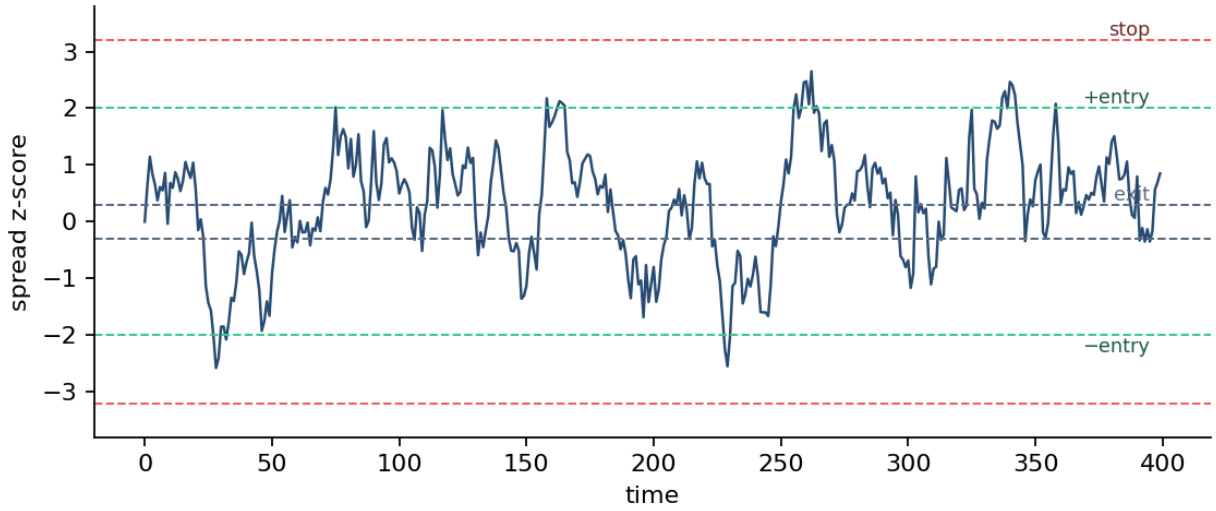


Figure 3 — Spread z-score bands on a synthetic mean-reverting spread (half-life ≈ 11 bars). Entry at ±2, exit at ±0.3, stop at ±3.2.

Fee and risk discipline come straight from the autopsy: at a 20 bps round trip, only spreads whose expected capture exceeds fees plus slippage by a stated margin are admissible; sizing is a function of half-life (faster reversion, smaller holding, wider band); and nothing reaches a live enablement without a human decision — Rule A6 makes the promotion gate explicit. The first evaluation vehicle is the Stage-1 path-level replay over the losing-model windows: the same harness that tests the repair hypotheses tests correction trades, on the same data, with the same acceptance math (purged/embargoed CV, deflated Sharpe, per docs/solvers.md).

6 Milestones

Table 3 — Staging. Each milestone justifies the next.

milestone	content	acceptance
M0 — this memo	design agreed on the channel	AGREE from both lanes; human nod
M1 — pair screen offline	cointegration screen over perp universe from existing klines	ranked pair list + half-lives, reproducible script
M2 — minting job + viewer toggle	corr_* job; four predicates; graph renders typed edges	edges versioned; viewer shows provenance
M3 — replay evaluation	correction trades replayed on Stage-1 windows	expectancy net of fees, CIs, deflated Sharpe
M4 — proposal	ranked trade proposals to the human	human decision per Rule A6; nothing auto-enabled

Open questions to think about: whether funding-rate dislocations (already in every perp's meta) should mint their own edge predicate or ride trades_with provenance; whether same_sector should come from the seeded category tags, from an embedding of coin descriptions, or both; and how aggressively the neighbourhood should generalise beyond two legs before the residual stops being hedgeable.

References

[1] Engle, R. F., & Granger, C. W. J. (1987). Co-integration and error correction: Representation, estimation, and testing. *Econometrica*, *55*(2), 251–276.

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[3] Gatev, E., Goetzmann, W. N., & Rouwenhorst, K. G. (2006). Pairs trading: Performance of a relative-value arbitrage rule. *The Review of Financial Studies*, *19*(3), 797–827.

[4] Vidyamurthy, G. (2004). *Pairs trading: Quantitative methods and analysis.* Wiley.

[5] Avellaneda, M., & Lee, J.-H. (2010). Statistical arbitrage in the US equities market. *Quantitative Finance*, *10*(7), 761–782.

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